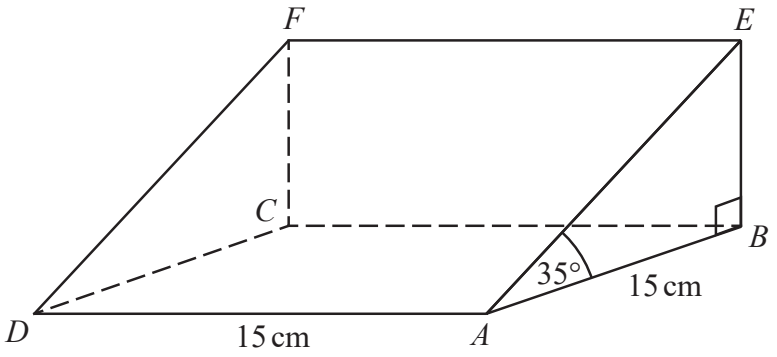
19 The diagram shows a triangular prism.



The base, $ABCD$, of the prism is a square of side length 15 cm.
Angle ABE and angle CBE are right angles.
Angle $EAB = 35°$

M is the point on DA such that

$$DM : MA = 2 : 3$$

Calculate the size of the angle between EM and the base of the prism.
Give your answer correct to 1 decimal place.

MA = 3/5 × 15 = 9 cm.

1 mark
Find MA or BE.

BE = 15 tan 35° ≈ 10.503 cm.

MB = √(15² + 9²) = √306 ≈ 17.493 cm.

1 mark
Pythagoras in base AMB.

The angle is EMB, above base MB.

tan(angle EMB) ≈ 10.503/17.493

1 mark
Use tan in right triangle MBE.

angle EMB ≈ 30.98° = 31.0° (1 d.p.)

1 mark
Angle 31.0°.

31.0 °

(Total for Question 19 is 4 marks)